MIL-B-2029F
25 February 1983
SUPERSEDING
MIL-B-2029E
23 November 1976

MILITARY SPECIFICATION

BURNER ASSEMBLY, SPACE HEATER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a liquid fuel burner assembly for a space heater.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766	-	Steel Plates, Sheet and Strip - Corrosion-Resisting
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-10868	-	Hose Assembly, Rubber, Utility, Gasoline (1/4 Inch Inside Diameter)
MIL-C-10872	-	Cap, Flue, 4-Inch, Cylindrical
MIL-A-10957	-	Adapter Kit, Gravity Feed for 5 Gallon Military Gasoline Can and 55 Gallon Drum
MIL-C-43343	-	Control, Fuel Flow, Oil Burner

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-11-975	-	Burner Assembly, Space Heater, Burner, Oil, Vaporizing Pot Assembly
5-11-976	-	Burner Assembly, Space Heater, Pot Assembly
5-11-977	-	Burner Assembly, Space Heater, Pot Details
5-11-978	-	Burner Assembly, Space Heater, Pilot Ring and Flame Spreader Assemblies
5-11-709	-	Burner Assembly, Space Heater, Instruction Sheet

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERALS

A 48	-	Gray Iron Castings
A 108	-	Steel, Bars, Carbon
A 109	-	Steel, Carbon, Cold-Rolled Strip
A 366	-	Steel, Carbon, Cold-Rolled Sheet, Commercial Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials and components. Materials and components shall conform to the applicable documents and requirements specified herein and on the applicable drawings. Materials and components not specifically defined shall be of the quality normally used provided that the completed item complies with all provisions of this document (see 6.4).

3.2.1 Gray cast iron. Gray cast iron shall conform to ASTM A 48.

3.2.2 Steel.

3.2.2.1 Carbon steel, sheet and strip. Sheet shall conform to ASTM A 366 and strip shall conform to ASTM A 109.

3.2.2.2 Carbon steel bars. Carbon steel bars shall conform to 1212 or 1213 composition of ASTM A 108.

3.2.2.3 Corrosion-resisting steel, sheet and strip. Corrosion-resisting steel shall conform to 304, 410 or 430 composition of QQ-S-766.

3.2.3 Thread compound. The thread compound for pipe joints shall be a commercial grade epoxy resin.

3.2.4 Paper. The paper for the instruction sheet shall be of chemical wood pulp stock, having a minimum weight of 90 pounds (per 500 sheets 22-1/2 by 28-1/2).

3.2.5 Pipe reducer. The pipe reducer shall be galvanized commercial grade.

3.2.6 Nipple. The nipple shall be galvanized commercial grade.

3.3 Design and construction. The burner assembly shall conform to the requirements specified herein and shown on the drawings listed in 2.1. In addition, the burner assembly shall consist of the components listed in table I.

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TABLE I. Burner assembly components

Quantity	Nomenclature
1	Burner, Oil, Vaporizing Pot, with Flame Spreader Consisting of: 1 Pot Assembly 1 Flame Spreader 1 Ring, Pilot, Burner, Oil, Heater, Space 1 Ring, Combustion, Burner; Oil, Heater, Space
1	Control, Fuel Flow, Oil Burner
1	Installation and Operating Instruction Sheet
1	Adapter Kit, Gravity Feed, Military Gasoline Can
2	Hose Assembly (see 3.4.5), Rubber, Utility, Gasoline, 1/4-Inch Inside Diameter by 12 Feet Long
1	Cap, Flue, 4-Inch, Cylindrical

3.4 Details of components.

3.4.1 Pot burner assembly. The pot burner assembly shall consist of the components shown on Drawing 5-11-975 and specified herein.

3.4.1.1 Pot base assembly. The pot base assembly (pot base, lower air tube, connector, and pot top) as shown on Drawing 5-11-976 shall not leak when tested as specified in 4.4.2.2.

3.4.1.2 Pipe reducer, nipple and pot base assembly. The pipe reducer, nipple and pot base assembly as shown on Drawing 5-11-976 shall be assembled with compound specified in 3.2.3 with no material obstruction to fuel flow.

3.4.1.3 Flame spreader. The flame spreader (upper air tube and chimney) shall be constructed as shown on Drawing 5-11-978. The upper air tube shall be free from blow holes, form distortions and material obstructions to air flow.

3.4.1.4 Pilot ring. The pilot ring shall be constructed as shown on Drawing 5-11-978.

3.4.1.5 Combustion ring. The combustion ring shall be constructed as shown on Drawing 5-11-978.

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3.4.2 Fuel flow control valve. The fuel flow control valve shall conform to MIL-C-43343.

3.4.3 Installation and operating instruction sheet. The installation and operating instruction sheet shall be printed in black lettering in accordance with Drawing 5-11-709, on an 8 by 10-1/2 inch paper sheet specified in 3.2.4. Installation instructions with burner assembly schematic shall be printed on one side and operating instructions on the reverse side. The instructions shall be neat, complete and legible.

3.4.4 Gasoline can adapter kit. The adapter kit shall conform to MIL-A-10957.

3.4.5 Rubber hose assembly. The rubber hose assembly shall be 12 feet long conforming to MIL-H-10868.

3.4.6 Flue cap. The flue cap shall conform to MIL-C-10872.

3.5 Identification marking. The letters US, the name of the prime contractor, contract number and year of contract shall be indented in the size and location shown on Drawing 5-11-977. When the manufacturer of the burner pot assembly is other than the prime contractor, this marking shall be supplemented by "MADE BY" followed by the name of the manufacturer. The words "THIS SIDE UP" shall be indented on the pilot and combustion rings as shown on Drawing 5-11-978. All letter indentations shall be legible with no evidence of penetration through metal or distortion of metal surfaces on opposite side.

3.6 Workmanship. The burner assembly components shall be clean, not sprung, broken or malformed and free from rust areas, burrs, slivers, bends, dents, creases, punctures, and fractures. All moving parts shall function without application of abnormal force. Screw threads shall not be marred, crossed or stripped.

3.6.1 Welding and brazing. All welding and brazing shall be continuous adherent, smooth, and free from scale or flux deposits. All spots welds or continuous welding shall be adherent and free from burn-through of metal or spatter.

3.6.2 Rivets. All rivets on the burner pot assembly shall be tight, fully peened and shall not be broken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First artcle inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and 4.4.3.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Visual examination of components.

4.4.2.1.1 Pipe reducer, nipple and pot base assembly. The pipe reducer, nipple and pot base assembly shall be examined for material obstruction to fuel flow. Any nonconformance to the requirements of 3.4.1.2 shall be considered a defect. The lot shall be all assemblies offered for inspection at one time. The sample unit shall be one assembly of the pipe reducer, nipple and pot base. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.2.1.2 Flame spreader, upper air tube. The flame spreader of upper air tube shall be examined for blow holes, form distortion and material obstruction to flow. Any nonconformance with requirements of 3.4.1.3 shall be considered a defect. The lot shall be all flame spreaders of the upper air tube offered for inspection at one time. The sample unit shall be one flame spreader of the upper air tube. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.2.1.3 Pot base assembly. The pot base assembly (pot base, lower air tube, connector and pot top) shall be examined. Any nonconformance with the welding requirements of 3.6.1 shall be considered a defect. The lot shall be all pot base assemblies offered for inspection at one time. The sample unit shall be one pot base assembly. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.2.2 Pot base assembly leak testing. Each pot base assembly, (pot base, lower air tube, connector and pot top) shall be tested as follows for conformance to the leakage requirement in 3.4.1.1. The lower portion of the pot assembly shall be sealed to a point below the lower row of holes of the pot top and above the weld joining the pot top and pot base (approximately 5-3/4 inches below the pot top).

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The lower air tube and the connector, nipple or the reducer shall also be sealed. The sealed pot base assembly shall be pressurized with air at 10 pounds per square inch and submerged in a clear liquid and examined for leaks. Any evidence of leakage shall constitute failure of the test and shall be cause for rejection of the pot base assembly.

4.4.3 End item inspection. The lot shall be all complete pot burner assemblies offered for inspection at one time. The sample unit shall be a complete pot burner assembly.

4.4.3.1 Visual examination. The pot burner assembly shall be examined for defects in table II. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Construction and workmanship (applicable to all components)	Part missing or not as specified	X	
	Fractured, broken, punctured, torn, dented, creased, bent, sprung, or malformed	X	
	Burrs or slivers		X
	Not connected or joined as specified	X	
	Not clean and not free from rust areas		X
	Binding of moving parts		X
Welding	Welding and brazing not continuous, adherent or smooth	X	
	Not free of scale or flux deposits		X
	Spot welds not adherent and free from burn-through of metal	X	
	Not free from spatter		X
Rivets	Missing, not type specified, broken	X	
	Loose or not fully peened		X
Identification marking	Missing, incomplete, not legible		X
	Indentation excessive with metal surface distortion		X
	Indentation excessive with metal penetration	X	
Instructions	Instruction sheet dirty, or lettering smudged		X
	Missing, torn, not legible, or incomplete	X	

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4.4.3.2 Dimensional examination. Components of the burner assembly shall be examined for nonconformance in dimensions. Any dimension not within specified tolerances shall constitute a defect. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with section 5 requirements. Defects shall be scored as listed in table III. The sample unit shall be one shipping container fully packaged except that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. Packaging defects

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, inadequate stapling. Bulged or distorted container.
Weight	Weight of contents per shipping container exceeds specified limit.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The combustion ring, pilot ring, and upper air tube shall be disassembled from the burner. The upper and lower vent tubes shall be disassembled from the adapter.

5.1.1.2 Application of preservative. All uncoated ferrous metal surfaces of the burner components and accessories, except the fuel control valve, shall be cleaned process C-1 and thoroughly dried, and coated with type P-7 preservative, in accordance with MIL-P-116. The reducer on the end of the 3/8-inch burner nipple and open ends of the fuel hose shall be sealed with non-corrosive type plugs or caps or tape conforming to type III or IV of PPP-T-60.

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5.1.1.3 Unit packing.

5.1.1.3.1 Cap, flue. The flue cap shall be preserved and wrapped as specified in MIL-C-10872, except that the surplus greaseproof barrier material at the open end of the flue cap shall be tucked inside.

5.1.1.3.2 Adapter kit and fuel control valve. The adapter kit and fuel control valve shall be individually wrapped in greaseproof barrier material conforming to type I or II, grade A, class 2 of MIL-B-121, and secured with 1-1/2-inch minimum width tape conforming to type III, grade B of PPP-T-45.

5.1.1.3.3 Instruction sheet. Each instruction sheet shall be prepared method IC-1 of MIL-P-116 in a flat position without folding or creasing.

5.1.1.3.4 Components, and accessories. The components, and disassembled accessories, preserved as specified, shall be placed within the pot in the following manner: The mouth of the flue cap shall be placed over the lower air tube. The upper air tube (large end down), the fuel control valve, and the adapter kit shall be placed in the most compact manner around the flue cap. The two lengths of hose shall be coiled around the inside of the pot. The pilot ring shall be placed on top of the contents in the pot, and the combustion ring shall then be placed in its normal position and secured with the four screws provided for that purpose. The burner shall then be completely wrapped in greaseproof barrier material conforming to type I or II, grade A, class 2 of MIL-B-121, and secured with 1-1/2 inch minimum width tape conforming to type III, grade B of PPP-T-45.

5.1.1.3.5 Burner. Each complete burner, with components, prepared as specified in 5.1.1.3.4, shall be unit packed upright with reducer in a corner, in a snug-fitting fiberboard box conforming to style RSC, W5c or W5s of PPP-B-636. Fiberboard pads, made from the same material as the box, shall be provided as required to protect the nipple and reducer. The installation and operating instruction sheet prepared as specified in 5.1.1.3.3 shall then be placed flat on top of the wrapped burner. Each box shall be closed in accordance with method IV as specified in appendix of PPP-B-636.

5.1.2 Commercial. Burners shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Burners, preserved as specified in 5.1, shall be packed in a snug-fitting cleated plywood or nailed wood shipping container conforming to overseas type 2 load of PPP-B-601, or class 2, style 2 or 4, type 2 load of PPP-B-621. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification. Weight of contents per shipping container shall not exceed 150 pounds.

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5.2.2 Level B packing. Burners preserved as specified in 5.1, shall be packed in a snug-fitting cleated plywood or nailed wood shipping container conforming to domestic type, style A or B of PPP-B-601 or class 1, style 2 or 4 of PPP-B-621. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification. Weight of contents per shipping container shall not exceed 150 pounds.

5.2.3 Commercial packing. Burners, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The burner assembly is designed, primarily, to burn automotive combat gasoline in Heater, Space, Radiant Type, Coal or Oil, MIL-H-13514. This burner assembly is also capable of burning other fuels when approved, such as kerosene, light fuel and light arctic diesel oil.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) When a first article is required (see 3.1, 4.3 and 6.3).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one burner. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Changes from previous issue. Asterisks are not used to identify changes with respect to the previous issue due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-2029F Burner Assembly, Space Heater

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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